Tutorial of energy minimization for molecules by MolAICal

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1. Introduction

Sometimes, the docked ligands in the pocket of receptors appear slightly conformationally distorted, and some researchers need to optimize the structures of ligands for some purpose. The molecular energy minimization can provide a good way to optimize the molecules into more reasonable state. Here, MolAICal supplies two ways for molecular energy minimization: one is the minimization implementation by MolAICal, the other is based on the interface of MolAICal and UCSF Chimera. For more introduction to energy minimization for molecules, please see the manual of MolAICal.

2. Materials

2.1. Software requirements

1) MolAICal: https://molaical.github.io or https://molaical.gitlab.io
2) UCSF Chimera: https://www.cgl.ucsf.edu/chimera
Notice: Make sure the installation of MolAICal is correct!

2.2. Example files

1) All the necessary tutorial files are downloaded from:
https://github.com/MolAICal/tutorials/tree/master/025-minimization

This tutorial presents two distinct approaches for energy minimization, detailed in Part I and Part II, respectively. Readers may choose to consult **either section individually** or both sequentially, depending on their requirements.

Part I. Energy minimization by MolAICal

Go into the folder "025-minimization", which contains some protein files (GCGRH.mol2 and GCGRH.pdb), ligand files (lig_10.mol2 and lig_10.pdb), and the files including the character "_fix" to assign the fixed atoms for energy minimization.

Check one or two molecule information for ensuring atom serials of the inputted molecules:
```
#> molaical.exe -min1 -i lig_10.mol2
#> molaical.exe -min2 -m info -i1 GCGRH.pdb -i2 lig_10.pdb
```

For minimizing one molecule:
1) Minimize a single molecule without the fixed atoms
```
#> molaical.exe -min1 -i lig_10.mol2 -o min_lig.mol2
```

2) Minimize a single molecule with the serial number of fixed atoms
```
#> molaical.exe -min1 -i lig_10.mol2 -o min_lig.mol2 -f 1-2
```

3) Minimize a single molecule with the reference fixed molecule file, iteration numbers of quick optimization and refinement

#> molaical.exe -min1 -i lig_10.mol2 -o min_lig.mol2 -f lig_fix.mol2 -q 10 -r 6

Minimize two molecules, output only the result of the second input molecule:

1) Minimize two input molecules

#> molaical.exe -min2 -m 1 -i1 GCGRH.pdb -i2 lig_10.pdb -f1 GCGRH_fix.pdb -f2 0 -o2 min_lig_1.pdb

2) Convert the second input PDB molecule in Mol2 and minimize two input molecules

#> molaical.exe -min2 -m 1 -i1 GCGRH.pdb -i2 lig_10.pdb -f1 GCGRH_fix.pdb -f2 0 -o2 min_lig_2.pdb -c2

Analysis: If "-f2" is 0, it means no fixed atoms the second molecule. Open ligands: lig_10.pdb, min_lig_1.pdb, and min_lig_2.pdb. Because the molecule in **PDB format has no bond connectivity information, some bonds will be changed** when energy minimization is performed. Please see Figure 1. **The molecule in mol2 format contains bond connection information, so it has the bond missing issue.** For fixed atoms, bond connectivity remains unchanged; thus, the fixed PDB-format molecules can be directly used for energy minimization without bond parameter consideration. "-c2" option converts the second molecule (in PDB format) to mol2 format and then performs energy minimization on the converted structure.

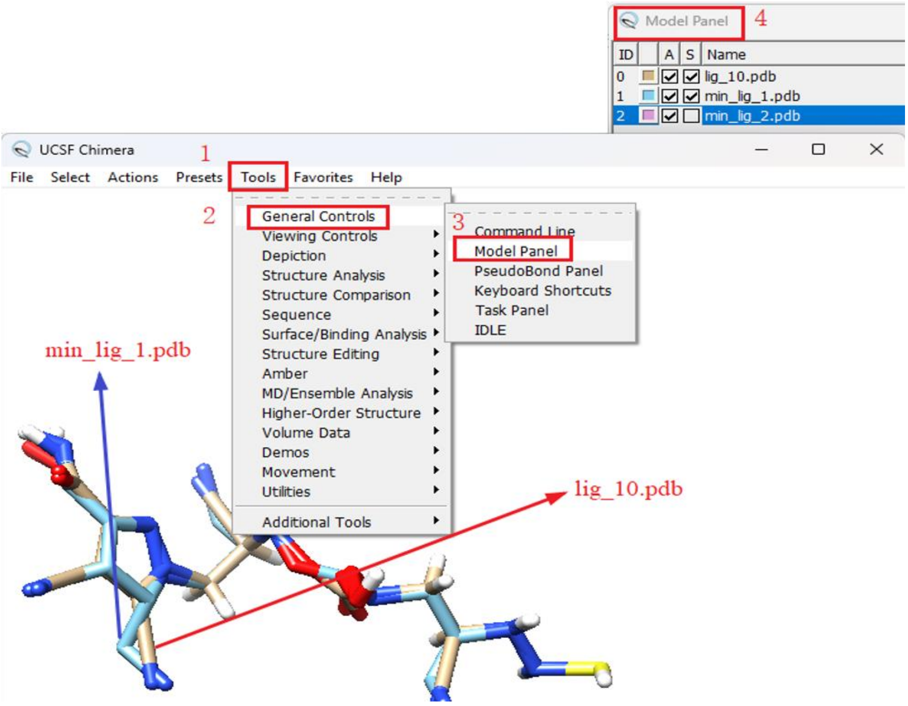

Figure 1

Minimize two molecules, output results for both input molecules:

1) Minimize two input molecules

#> molaical.exe -min2 -m 2 -i1 GCGRH.pdb -i2 lig_10.pdb -o1 min_gcgr.mol2 -o2 min_lig.mol2

2) Convert the second input PDB molecule in Mol2 and minimize two input molecules
#> molaical.exe -min2 -m 2 -i1 GCGRH.pdb -i2 lig_10.pdb -o1 min_gcgr.mol2 -o2 min_lig.mol2 -c2

For the parameters of minimization of MolAICal, please see the content below:

-f1 <arg>	Either fixed atoms (e.g., '1-3,5,7-9') or reference molecule file for first molecule.
-f2 <arg>	Either fixed atoms (e.g., '1-3,5,7-9') or reference molecule file for second molecule.
-h	Print help information.
-i1 <arg>	Input molecule file for first molecule (.pdb, .mol2, .sdf, .mol).
-i2 <arg>	Input molecule file for second molecule (.pdb, .mol2, .sdf, .mol).
-m <arg>	Operation mode for minimization (default: null), value can be **info, 1, or 2**: “**info**” means to print molecules' information only; “**1**” means to minimize one molecule only; “**2**” means to minimize two molecules.
-o1 <arg>	Output file for minimized molecule for first molecule (.pdb, .mol2, .sdf, .mol).
-o2 <arg>	Output file for minimized molecule for second molecule (.pdb, .mol2, .sdf, .mol).
-q <arg>	The iteration number of Quick optimization.
-r <arg>	The iteration number of Refinement.
-c2	Convert the second input PDB file in mol2 format if "-c" in the command line, it is designed for molecular docking of MolAICal.

Part II. Energy minimization by UCSF Chimera and MolAICal

Should energy minimization in Part I suffice for the research objectives, this section may be omitted. In cases where the Part I procedure yields erroneous results, Part II serves as a validated alternative methodology. Go into the folder “025-minimization”:

Mode 0: It can run the script file of UCSF Chimera directly

1) Run the command script of UCSF Chimera by MolAICal

#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 0 -i1 run.com

Note: “-c” is needed to point to the path of the program “chimera”, “run.com” is the command script file of UCSF Chimera.

Mode 1: minimization for one molecule

1) Minimize one molecule with the assigned atom serials

#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 1 -i1 lig_10.mol2 -sv "#0:@/serialNumber>=20" -o1 out_1.mol2

Note:

- ✧ Here, the option "-i1" is the input molecular file, which is different from Mode 0. The option "-sv" is used to assign the fixed atoms to perform minimization. The selected ways for the fixed atoms are employed from the styles of UCSF Chimera. For more detailed information, please see https://www.rbvi.ucsf.edu/chimera/current/docs/UsersGuide/midas/frameatom_spec.html
- ✧ For Modes 1 and 2, the value following "-sv" can be: "**none**", "**selected**", "**unselected**", or **atom specifications** (atom-spec) using UCSF Chimera's atom selection syntax.

Mode 2: minimization for two molecules

1) Minimize two molecules with the assigned atom specifications and output two molecules

#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 2 -i1 lig_10.mol2 -i2 GCGRH.pdb -sv "#2:<0>" -os "#2:<0>" -o1 out_1.mol2 -o2 out_2.pdb

Note:

- ✧ The option "-i1 lig_10.mol2" corresponds to '**#0**';
- ✧ The option "-i2 GCGRH.pdb" corresponds to '**#1**';
- ✧ The option -sv "#2:<0>" specifies selecting and fixing the residue named "<0>" in the merged complex (#2) for subsequent energy minimization. Here, #2 refers to the complex formed by merging GCGRH.pdb (#1) and lig_10.mol2 (#0).
- ✧ For Modes 1 and 2, the value following "-sv" can be: "**none**", "**selected**", "**unselected**", or **atom specifications** (atom-spec) using UCSF Chimera's atom selection syntax.
- ✧ The option -os "#2:<0>" is used to select one molecule (<0>) from the complex (#2) for saving after energy minimization, while the other molecule is saved by excluding the selected one. This is because energy minimization is performed on the merged complex (#2) of the first molecule (#0) and the second molecule (#1).

Mode 3: minimization for two molecules with one fixed molecule

1) Minimize two molecules with one fixed molecule and output two molecules

#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 3 -i1 lig_10.mol2 -i2 GCGRH.pdb -f1 GCGRH_fix.pdb -sv "#0 za<0.01" -os "#3:<0>" -o1 out_1.mol2 -o2 out_2.pdb

Note:

- ✧ For Modes 3 and 4, the value following -sv must be an atom-spec (the atom selection syntax in UCSF Chimera) and **zone specifiers**.
- ✧ In Mode 3, the default value of -sv is "#0 za<0.01".
- ✧ **Zone specifiers** define atoms and residues located within or beyond a specified distance from reference atom(s). The operators z< and zr< select all residues containing any atom within the given distance from reference atoms, while za< selects all individual atoms within that distance. The complementary operators z>, zr>, and za> return the inverse sets of their < counterparts.

  For example:

  #1:UTK za<12.5

  Means to select all atoms within 12.5 Å of any atom in residue UTK from model #1.

Mode 4: minimization for two molecules with two fixed molecules

1) Minimize two molecules with two fixed molecules, output one molecule and log file based on the full options of MolAICal

#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 4 -i1 lig_10.mol2 -i2 GCGRH.pdb -f1 GCGRH_fix.pdb -f2 lig_fix.mol2 -os "#4:<0>" -cm gas -fs freeze -sv "#0,1 za<0.01" -fv false -pv false -ns 10 -ss 0.02 -cs 2 -css 0.02 -iv 1 -ng true -mf1 mol2 -mf2 pdb -o1 out_1.mol2 -o2 out_2.pdb -w false -p false

Note:

- For Modes 3 and 4, the value following -sv must be an atom-spec (the atom selection syntax in UCSF Chimera) and **zone specifiers**.
- In Mode 4, the default value of -sv is "#0,1 za<0.01".
- **Zone specifiers** define atoms and residues located within or beyond a specified distance from reference atom(s). The operators z< and zr< select all residues containing any atom within the given distance from reference atoms, while za< selects all individual atoms within that distance. The complementary operators z>, zr>, and za> return the inverse sets of their < counterparts.

  For example:

  #1:UTK za<12.5

  Means to select all atoms within 12.5 Å of any atom in residue UTK from model #1.

For the parameters of minimization of UCSF Chimera and MolAICal, please see the content below:

-c,--chimera_path <arg> Path to Chimera executable

Notice: MolAICal also provides a method to set the Chimera path, Please note that the configuration of MolAICal differs between the Windows and Linux versions: **the Linux version utilizes the udocker container approach and requires setup within the container**, whereas the Windows version does not. For the configuration or installation of UCSF Chimera in the Linux version of MolAICal, users can refer to the NAMD installation steps in **Appendix 1.** The final step of configuring UCSF Chimera is very similar in both Linux and Windows versions of MolAICal, as follows:

```
#> molaical.exe -call set -n chimera -p "d:\Program Files\Chimera119\bin\chimera.exe"
```

The value of '-n' **must** be **'chimera'** (case-insensitive). Currently, **'VMD', 'NAMD'**, and **'chimera'** are fixed names for internal path specifications in **MolAICal**. Therefore, they must be written exactly as shown, though letter case does not matter. '-p' is the absolute path of the executable program of UCSF Chimera.

Once set, this path will be saved and used automatically, **eliminating the need to input chimera_path every time**, thus reducing repetitive input. **Thus, it is not necessary to input '-c' or '--chimera_path' when performing minimization.** For the above example of Mode 3, the command can be input as follows:

```
#> molaical.exe -cmin -m 3 -i1 lig_10.mol2 -i2 GCGRH.pdb -f1 GCGRH_fix.pdb -sv "#0 za<0.01" -os "#3:<0>" -o1 out_1.mol2 -o2 out_2.pdb
```

-cm,--charge_method <arg>	Charge method (am1 or gas), default: gas
-cmin	minimization calculation from the interface for UCSF Chimera.
-cs,--cgsteps_value <arg>	Conjugate Gradient (CG) steps, default: 2
-css,--cgstepsize_value <arg>	Conjugate Gradient (CG) step size, default: 0.02
-f1,--fixed_f1 <arg>	Path to first fixed molecule file (for modes 3 and 4)
-f2,--fixed_f2 <arg>	Path to second fixed molecule file (for mode 4)
-fs,--freeze_spec <arg>	Freeze or spec method, default: freeze
-fv,--fragment_value <arg>	Whether to use fragment (true/false), default: false
-h,--help	Display help information
-i1,--input_f1 <arg>	Path to first input molecule file or command file (for mode 0)
-i2,--input_f2 <arg>	Path to second input molecule file or output log (for mode 0)
-iv,--interval_value <arg>	Interval, default: 1
-m,--mode <arg>	Mode of operation (0, 1, 2, 3, or 4): **Mode 0**: Direct call to chimera_command (--input_f1 as commandFile, --input_f2 as outputLog); **Mode 1:** Single molecule minimization; **Mode 2:** Two-molecule minimization; **Mode 3:** Three-molecule minimization with one fixed; **Mode 4:** Four-molecule minimization with two fixed.
-mf1,--mol_format_1 <arg>	First output format, default: mol2
-mf2,--mol_format_2 <arg>	Second output format, default: pdb
-ng,--nogui_value <arg>	Whether to use GUI (true/false), default: true
-ns,--nsteps_value <arg>	Steepest descent steps, default: 10
-o1,--output_f1 <arg>	First output file
-o2,--output_f2 <arg>	Second output file
-os,--output_sel <arg>	Output selection
-p,--print_output <arg>	Whether to print output (true/false), false means to output the log file, vice versa, default: true
-pv,--prep_value <arg>	Whether to use prep (true/false), default: true
-ss,--stepsize_value <arg>	Steepest descent step size, default: 0.02
-sv,--sel_value <arg>	Selection value for atoms
-w,--whether_output <arg>	Whether to output the second molecule (true/false), false means no second output, vice versa, default: true

Appendix 1

Install external programs inside the MolAICal container (if finished, only for Linux Version MolAICal)

Please note that the configuration of MolAICal differs between the Windows and Linux versions: the Linux version utilizes the udocker container approach and requires setup within the container, whereas the Windows version does not.

To address the library dependency issues in Linux, the Linux version of MolAICal adopts a container-based approach. If **external programs need** to be invoked, it is **recommended** to install them **within** the MolAICal **container**. If **no external** programs are required, this step can be **ignored**. This tutorial requires the external programs NAMD and VMD, and the steps are as follows:

a) First of all, copy the files to the MolAICal container.

```
****** 1.   Go to the container file system, it will enter the "/root" ******
#> molaical.exe -eset shell in

****** 2.   The first part of 'cp' is from the local machine, and the second part is in the container; ******
****** The VMD and NAMD packages can be moved into the container by the command below ******
#> cp /home/user/<local machine file> /root/soft

****** 3.   Exit container file system ******
#> exit
```

b) Secondly, enter the container virtual environment of MolAICal:

```
#> molaical.exe -eset sys run molaical
```

Note: 'molaical' is the container name.

c) Installing software in the container virtual environment of MolAICal is the same as installing it on the host local machine. Here, we take the installation of VMD and NAMD as examples:

1) For NAMD:

Extract the NAMD files, assuming the extracted folder is named namdcpu. Then specify its path to MolAICal using the following command:

```
#> molaical.exe -call set -n NAMD -p "/root/soft/namdcpu/namd3"
```

Note: Please **replace** the above paths for VMD and NAMD with the **actual paths** on the users' system. The string "VMD" and "NAMD" (case insensitive) after "-n" is fixed for the paths of VMD and NAMD. To ensure the reproducibility of MM/GBSA results, it is recommended to use the CPU version of NAMD, as the "seed" parameter appears to be ineffective for reproducibility in the CUDA version of NAMD.

2) For VMD:

Following the steps below:

✧ Extract the VMD files:

```
#> tar -xzvf vmd-xxx.tar.gz
```

Note: Please **replace** the above paths for VMD and NAMD with the **actual paths** on the users' system.

✧ Modify the install path in a file named "**configure**" in the decompressed folder of VMD:

```
The default: $install_bin_dir="/usr/local/bin"; modify it to →
$install_bin_dir="/root/soft/vmd193/bin";

The default: $install_library_dir="/usr/local/lib/$install_name"; modify it to →
$install_library_dir="/root/soft/vmd193/lib/$install_name";
```

✧ Install VMD:

```
#> cd vmd-xx
#> ./configure LINUXAMD64
#> cd src
#> make install
```

Note: Please run **./configure**, and select the right types for the used computers. Here, it is "LINUXAMD64".

✧ Then specify its path to MolAICal using the following command:

```
#> molaical.exe -call set -n VMD -p "/root/soft/vmd193/bin/vmd"
```

So far, all the installations and configurations of NAMD and VMD are complete in the MolAICal container.

3) Save and load the modified container (Option)

To preserve changes made within a container and avoid reinstalling software later, since all data will be lost if the container is deleted.

◆ The first step is to obtain the container ID and name:

```
#> molaical.exe -eset sys ps
```

◆ Secondly, clone this container as follows:

```
#> molaical.exe -eset sys export --clone -o molaicalv2.tar molaical
```

Note: 'molaical' is the container name. It can also be the container ID. If an error is reported, it may be due to permission issues with the mounted file system, which can be ignored.

◆ Thirdly, import the cloned container **if needing**:

```
#> molaical.exe -eset sys import --clone --name molaicalv2 molaicalv2.tar
```

◆ Now, change the loaded new container name to the default container name "molaical", the previous container is renamed to "bakmolaical", the commands are as follows:

```
#> molaical.exe -eset sys rename molaical bakmolaical
#>molaical.exe -eset sys rename molaicalv2 molaical
```

Notice: Please note that a **container** (**dynamic**) is generated from an **image** (**static**). Operations such as software installation must be performed within the dynamic container; if cloning this container, restoring it will still be based on the original underlying image.

For more information, please see the manual of MolAICal or run "molaical.exe -eset sys --help" to get more help details.